

Math 340-004: Elementary Matrix and Linear Algebra

MIDTERM TWO (Note: This is a typo, and should be **Final Exam**)

May 5, 2019

TABLE OF DISCUSSION SECTIONS		Section	Meets	Leader
361	M		8:50 - 9:40 am	Ongay Valverde
362	M		9:55 - 10:45 am	Ongay Valverde
363	M		11:00-11:50 am	Mohamed
364	M		12:05-12:55 pm	Mohamed
365	M		1:20-2:10 pm	Mohamed

Name: _____ Section#: _____ NetID: _____

I pledge that the work on this exam is entirely my own. I understand that the penalties for cheating may include an F in the course and referral to my dean for further action.

Student Signature: _____

READ THE FOLLOWING INFORMATION.

- This is a **90-minute** exam. It consists of 6 problems on 7 pages. It is your responsibility to make sure that you have a complete exam.
- ~~You may use a standard scientific calculator, but not one which has sizeable stored memory.~~
- Books, notes, and other aids are not allowed.
- You may use the backs of the pages or any additional pages for scratch work, but it will not be graded. ***Do not unstaple or remove pages as they can be lost in the grading process.***
- Only complete, well written, and neat solutions will be awarded full credit. Remember that all claims must be supported. Responses which do not meet these qualifications may be awarded some partial credit.

DO NOT BEGIN THIS EXAM UNTIL SIGNALLED TO DO SO.

1. (16 points) For each of the following, please give an appropriate definition for the word or phrase in *italics*.

(a) The set W is a *subspace* of a vector space V .

(b) The set S is a *basis* for a vector space V .

(c) The function L is a *linear transformation* from V to W .

(d) The a linear transformation has *rank* n .

(e) ~~The *geometric multiplicity* of an eigenvalue of the matrix A .~~

(f) The set C is the *null space* of a matrix A .

(g) A is a *singular* matrix.

(h) S is an *orthonormal* basis for the inner product space V .

2. (12 points) Let

$$A = \begin{bmatrix} 1 & 3 & -2 \\ 2 & 7 & -5 \\ 3 & 9 & x \end{bmatrix}$$

where x is some real number. Compute the following (if possible). If the computation is not possible, explain why.

(a) Row reduce A to an upper triangular matrix.

(b) What is the determinant of A ?

(c) Find all values of x which make the matrix singular.

(d) Find all values of x for which the rank of the matrix is 3.

3. (9 points) Let $L : P_3 \rightarrow P_3$ by

$$L(p) = p'(t) + p''(t).$$

(a) Let $\gamma = \{t^3, t^2, t, 1\}$ be the standard basis for P_3 . Give a matrix representation for L under this basis.

(b) Find a basis for $\ker L$.

(c) Find a basis for $\text{range } L$.

4. (12 points) Consider the inner product space $P_2[0, 1]$ of polynomials of degree 2 on the interval $[0, 1]$ with the inner product

$$(p, q) := \int_0^1 pq \, dx$$

- (a) Find a unit vector parallel to $p(t) = t$.

~~(b) Let $S = \text{span}\{1, t + 2\}$. Find an orthonormal basis for S .~~

- (c) With S as in part (b), ~~compute $S^\perp$.~~ Find a basis for the set of vectors orthogonal to S .

~~(d) With S as in part (b), find the vector v in S so that $\|v - x^2\|$ is as small as possible.~~

5. (6 points) Let

$$A = \begin{bmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(a) What is the characteristic polynomial of A ?

(b) Is A diagonalizable? Why or why not?

6. (16 points) Indicate (clearly) if the following statement is true or false. In either case give a short explanation why.

(a) A linear system of three equations can have exactly two solutions.

(b) A product of invertible matrices is non-singular.

(c) $\det(AB^T A^{-1}) = \det(B)$.

(d) A linearly independent set can have a linearly dependent subset.

(e) If v is orthogonal to both u and w then v is orthogonal to every vector in $\text{span}\{u, w\}$.

(f) There is a linear transform $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ which has a trivial kernel.

(g) There is a linear transform $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ which has trivial kernel.

(h) Every 2×2 matrix is invertible.